

Bilateral Tuning of Robust Speed Control for Permanent-Magnet Synchronous Motor System with Variations of Load Inertia and External Disturbance*

Stone CHENG** and Yi-Ming TENG**

** Department of Mechanical Engineering

National Chiao Tung University

1001 University Road, Hsinchu, Taiwan, ROC

E-mail: stonecheng@mail.nctu.edu.tw

Abstract

The dynamic performance of a permanent magnet synchronous motor (PMSM) is influenced practically by uncertainties in parameters including external load disturbance and inertia variation. This paper presents robust control to maintain dynamic performance of a PMSM with system inertia variation and external disturbance effects. The proposed algorithm used an extended Kalman filter (EKF) to identify system inertia accurately and then tuned the parameters of the speed controller. Furthermore, the estimated inertia is sent to adjust the reference model of the disturbance observer, and the estimated disturbance torque is fed forward to adjust the electromagnetic torque reference simultaneously. Both control units, system identification and disturbance observer, adaptively correct each other's estimated parameters. The proposed speed control scheme is implemented on a PMSM driver by a digital signal processor (DSP) platform, and the performance effectiveness is verified through MATLAB/Simulink simulations and experiments. To confirm its validity, the proposed speed controller is also compared with an industrial PMSM controller in real-time experiments under different practical conditions.

Key words: PMSM, Inertia Identification, Disturbance Observer, EKF

1. Introduction

Currently, permanent magnet synchronous motors (PMSMs) are often selected as actuators in many types of automatic industrial equipment such as CNC machining centers and milling machines. In practice, a PMSM servo driver generally regulates the best control gain for fixed loaded or unloaded conditions. However, the servo system is affected by uncertainties in inertia variation and external load disturbance, both of which seriously deteriorate system performance. For several industry applications, instantaneous machining disturbances such as backlash and stress inevitably exist. To compensate for the effect on speed response, the basic idea is to model a reference system and estimate the modeling error between the real system and the reference system to reduce the effect of external disturbances. The advantage of using an observer is that there is no need to consider the cause of disturbances or to model the nonlinearity of frictional disturbances.

In fact, most disturbances are caused by transmission mechanics, such as bearings, gears, ball screws, and the roller axis of a printing press, but the inertia can also change at the same time. Many disturbance-observer-based (DOB) control methods have been

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reported⁽¹⁾⁻⁽¹²⁾ which estimate mechanical inertia-influenced external disturbances, such as an extended Kalman filter (EKF)⁽¹⁾⁽²⁾, affine projection algorithm (APA)⁽³⁾, full-order state observer⁽⁴⁾⁽⁵⁾, adaptive torque observer⁽⁷⁾⁽⁸⁾, extended nonlinear observer⁽⁹⁾⁻⁽¹¹⁾, and extended state observer (ESO) controller⁽¹³⁾. However, the above methods ignore variations in system inertia that would also relate to the estimated external intrusion disturbance. It is important to interpret the disturbance observer as being sensitive to inertia variations. To achieve high performance for speed tracking and load disturbance compensation for speed stability, accurate values of mechanical inertia must be identified.

Several approaches are available for inertia identification. The method of state variable integration⁽¹⁴⁾⁽¹⁵⁾ is easy to realize, but it will induce a problem with the gain of the proportional-integral (PI) controller that will interfere with whether the correct value is estimated. A model reference adaptive system (MRAS)⁽¹⁶⁾ is sensitive to the adaptive gain, and the estimated inertia oscillates critically in different mechanical-parameter conditions. Recursive extended least square (RELS)⁽¹⁷⁾ is relevant only in the low-speed region. The reduced-order extended Luenberger observer (ROELO)⁽¹⁸⁾ is used in online inertia estimation, but overshoot still exists after inertia is identified. An ESO-based adaptive controller⁽¹⁹⁾ shows good load inertia identification, and the speed response has a much smaller overshoot.

In this paper, we propose disturbance estimation by a full-order state observer and system inertia identification based on an EKF. The EKF estimates the filtered state from the noised outputs provided from electromagnetic torque and motor velocity feedback to assure a good localization. To relax the burden of computation⁽¹⁾⁽²⁾, the EKF used in this study only calculates the system inertia from electromagnetic torque, mechanical angular speed, and the estimated disturbance torque. In fact, the equivalent disturbance estimated by the full-order state observer includes not only the actual external disturbances but also inertia-influenced disturbances. Nevertheless, the inertia-influenced disturbance can be used to identify inertia variation when the motor is accelerating or decelerating, and then the identified inertia bilaterally tunes the reference model of the disturbance observer and speed PI controller. Consequently, the actual external disturbance and the inertia-influenced disturbance would be separated from the estimated disturbance. Also, the estimated disturbance is fed forward to adjust the electromagnetic torque reference simultaneously.

The dominant parameters of the speed control loop are tuned automatically to promote dynamic response when inertia varies and an external disturbance occurs. This control scheme design is the main contribution of this paper. The remainder of this paper is organized as follows. In Section II, the PMSM model is described. In Section III, the speed control scheme with a full-order disturbance observer and system inertia identification is presented. The distinction of actual and inertia-influenced disturbance from the source of estimated disturbance is also developed. Simulation and experimental results will be presented in Sections IV and V, respectively. Finally, some conclusions are discussed in Section VI.

2. Problem description

Because the essence of a PMSM is its nonlinear properties, it is difficult to control the coupling magnetic flux. In 1970, Hasse and Blaschke proposed the vector control theory field-oriented control (FOC)⁽¹⁹⁾⁽²⁰⁾. It is widely used to simplify the AC servo motor driver as a linear system. The PMSM model at the rotor synchronous coordinate axis (d-q axis) could be expressed as (1). It can be separately controlled with the electromagnetic torque and rotor flux from the coupling magnetic flux. Therefore, by controlling the d-q axis current, it can be simply regarded as a DC motor model.

$$\begin{bmatrix} v_{ds} \\ v_{qs} \end{bmatrix} = \begin{bmatrix} r_s + sL_{ds} & -\omega_e L_{qs} \\ \omega_e L_{ds} & r_s + sL_{qs} \end{bmatrix} \begin{bmatrix} i_{ds} \\ i_{qs} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_e \lambda_f \end{bmatrix} \quad (1)$$

where ω_e , r_s , and λ_f are the electrical angular speed, stator resistance, and rotor flux linkage, respectively. i_{ds} and i_{qs} , v_{ds} and v_{qs} , and L_{ds} and L_{qs} are the stator current, voltage, and inductance, respectively.

The bandwidth of the PMSM current loop is generally higher than that of the speed loop; thus, the current controller can be approximated with a unity gain and be excluded from the mechanical model. The model of the PMSM mechanical system with an actual external load disturbance is expressed as (2) and shown in Fig. 1.

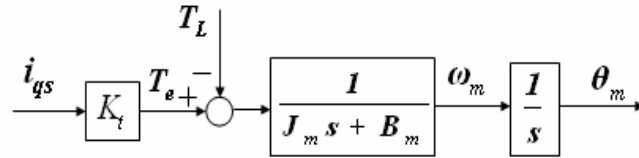


Fig. 1. The simplified model of PMSM mechanical system.

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases} \quad (2)$$

with

$$x = [\theta_m \quad \omega_m \quad T_L]^T, \quad u = T_e, \quad y = \theta_m$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -B_m/J_m & -1/J_m \\ 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1/J_m \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^T \quad (3)$$

where

θ_m is the angular position;
 ω_m is the mechanical angular speed;
 T_L is the external actual load;
 T_e is the electromagnetic torque;
 J_m is the rotor of inertia; and
 B_m is the friction coefficient.

3. Speed control strategy

A. Full-Order Disturbance Observer

In general, the time constant of the equivalent disturbance T_d is much larger than that of the speed controller because the sampling time is short in the speed loop. Under this assumption, we can consider the difference in the equivalent disturbance T_d to be zero in unit sampling time. We use the full-order state observer and choose the position, speed, and equivalent disturbance as state variables that can be arranged as the following equation.

$$\begin{bmatrix} \dot{\hat{\theta}}_m \\ \dot{\hat{\omega}}_m \\ \dot{\hat{T}}_d \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -B_m/J_m & -1/J_m \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{\theta}_m \\ \hat{\omega}_m \\ \hat{T}_d \end{bmatrix} + \begin{bmatrix} 0 \\ 1/J_m \\ 0 \end{bmatrix} T_e + K \left(\theta_m - [1 \quad 0 \quad 0] \begin{bmatrix} \hat{\theta}_m \\ \hat{\omega}_m \\ \hat{T}_d \end{bmatrix} \right) \quad (4)$$

where the symbols with “^” are estimated values. $\hat{T}_d$ is the estimated equivalent disturbance considering model uncertainty, and K is the observer gain matrix.

$$K = [k_1 \quad k_2 \quad k_3]^T \quad (5)$$

The characteristic version of (4) in the s -domain is given in (6) from the dynamic error equation, and the observer gains are chosen to make it converge stably.

$$\det[sI - (A - KC)] = 0 \quad (6)$$

The equation can be extended as

$$s^3 + \frac{k_1 J_m + B_m}{J_m} s^2 + \frac{k_2 J_m + k_1 B_m}{J_m} s - \frac{k_3}{J_m} = 0 \quad (7)$$

Suppose that the desired roots are placed at $\beta_1, \beta_2, \beta_3$ for the disturbance observer. The desired estimator characteristic equation can be denoted with $\alpha_c(s)$ as follows.

$$\alpha_c(s) = (s - \beta_1)(s - \beta_2)(s - \beta_3) = 0 \quad (8)$$

K can be solved by comparison with the coefficients in (7) and (8).

$$\begin{aligned} k_1 &= -(\beta_1 + \beta_2 + \beta_3) - B_m / J_m \\ k_2 &= (\beta_1 \beta_2 + \beta_2 \beta_3 + \beta_3 \beta_1) + (\beta_1 + \beta_2 + \beta_3) B_m / J_m + (B_m / J_m)^2 \\ k_3 &= \beta_1 \beta_2 \beta_3 J_m \end{aligned} \quad (9)$$

To analyze the estimated equivalent disturbance $\hat{T}_d$ by the observer, add the uncertainties of the parameters in (2).

$$(K_{Tn} + \Delta K_T)(i_{qsc} + \Delta i_{qs}) - T_L = (J_m + \Delta J_m) \frac{d\omega_m}{dt} + (B_m + \Delta B_m) \omega_m \quad (10)$$

where $T_e = K_{Tn} i_{qsc}$;

ΔK_T is the variation in the torque constant;

K_{Tn} is the nominal torque constant;

Δi_{qs} is the variation in the q -axis current;

i_{qsc} is the q -axis current reference;

ΔB_m is the variation in the friction coefficient; and

ΔJ_m is the variation in the rotor of inertia.

The model of the PMSM mechanical system with estimated equivalent disturbance $\hat{T}_d$ is expressed as follows.

$$J_m \frac{d\omega_m}{dt} + B_m \omega_m = T_e - \hat{T}_d \quad (11)$$

Substituting (11) in (10), the estimated equivalent disturbance $\hat{T}_d$ can be arranged as

$$\hat{T}_d = T_L - \Delta K_T i_{qsc} - K_{Tn} \Delta i_{qs} - \Delta K_T \Delta i_{qs} + \Delta B_m \omega_m + \Delta J_m \frac{d\omega_m}{dt} \quad (12)$$

Supposing the rotor flux and friction coefficient are constant and the deviation from the q -axis current is small, (12) can be simplified as:

$$\hat{T}_d = T_L + T_{\Delta J}, \text{ where } T_{\Delta J} = \Delta J_m \frac{d\omega_m}{dt} \quad (13)$$

The result of (13) means that the estimated equivalent disturbance $\hat{T}_d$ includes mainly the actual external load disturbance T_L and inertia-influenced disturbance $T_{\Delta J}$.

B. System Inertia Identification

The inertia of the PM servo motor system changes when the motor is joined with the load mechanism. The inertia of the two-mass system is related by the coefficients of elasticity and gear ratio, as shown in Fig. 2. This problem causes uncertainties in the reference model of the disturbance observer. From (10), we know that the variation in mechanical inertia degrades the transient response. Because a two-mass system is quite complex to analyze, we regard the servo motor system as a one-mass system and identify the equivalent inertia J_{sys} to maintain the dynamic response.

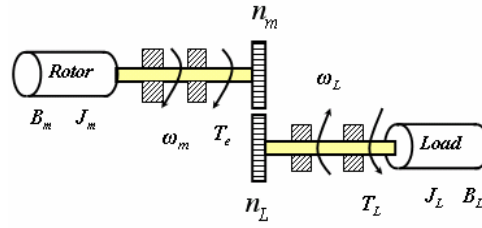


Fig. 2. Two –mass system.

We define J_{sys} as the equivalent system inertia. The moment of system inertia also changes slowly compared with the sampling time. Thus, the variation in the moment of inertia can be expressed as $\Delta J_{sys}/\Delta t=0$. The nonlinear difference equation can be expressed as follows.

$$\begin{cases} \frac{\omega_m[k+1]-\omega_m[k]}{T_s} = -\frac{B_m}{J_{sys}[k]}\omega_m[k] + \frac{T_e[k]-\hat{T}_d[k]}{J_{sys}[k]} \\ \frac{J_{sys}[k+1]-J_{sys}[k]}{T_s} = 0 \end{cases} \quad (14)$$

T_s is the sampling time, and the symbol and subscript “ k ” denotes the state variable at time k . From (14) the state variable equation can be written as

$$\begin{cases} x_{k+1} = x_k + f_k(x_k, u_k) + w_k \\ y_{k+1} = C_{k+1}x_{k+1} + v_{k+1} \end{cases} \quad (15)$$

where

$$f_k(x_k, u_k) = \begin{bmatrix} \frac{T_s}{J_{sys}[k]}(T_e[k]-\hat{T}_d[k]-B_m\omega_m[k]) \\ 0 \end{bmatrix} \quad (16)$$

$$x_k = [\omega_m[k] \quad J_{sys}[k]]^T, \quad C_k = [1 \quad 0] \quad (17)$$

x_k , y_k , $f_k(x_k, u_k)$, w_k , and v_k are the state variable vector, measurement variable vector, system model, system noise matrix, and measurement noise vector, respectively.

Both system and measurement noise are independent of each other. The system noise covariance Q_k is related to the minimum resolution of the estimated angular speed and system inertia. The measurement noise covariance R_k is used to eliminate the measurement noise from the sensor. Q_k and R_k are assumed to be constant and are defined as follows.

$$\begin{cases} cov(w_i, w_j) = Q_i \delta_{i,j} \\ cov(v_i, v_j) = R_i \delta_{i,j} \end{cases}, \quad \delta_{i,j} = \begin{cases} 1 & i \neq j \\ 0 & i = j \end{cases} \quad (18)$$

Because (14) is a nonlinear equation, it should be modified and linearized to the EKF for state estimation. First, $f_k(x_k, u_k)$ should be a partial differential of the state variables, written as follows.

$$F_k = \frac{\partial f(x_k)}{\partial x_k} = \begin{bmatrix} -\frac{B_m T_s}{J_{sys}[k]} & \frac{T_s}{J_{sys}^2[k]}(-T_e[k] + \hat{T}_d[k] + B_m \omega_m[k]) \\ 0 & 0 \end{bmatrix} \quad (19)$$

The EKF can be calculated by iteration as follows:

Step 1: Prediction of the state vector:

$$x_{k+1/k} = x_{k/k} + f_{x(k/k)} \quad (20)$$

Step 2: Estimation of error covariance matrix:

$$P_{k+1/k} = P_{k/k} + T_s (F_k P_{k/k} + P_{k/k} F_k^T) + Q_k \quad (21)$$

Step 3: Computation of Kalman gain:

$$K_{k+1} = P_{k+1/k} C_k^T (C_k P_{k+1/k} C_k^T + R_k)^{-1} \quad (22)$$

Step 4: Updating the error covariance matrix:

$$P_{k+1/k+1} = (I - K_{k+1} C_k) P_{k+1/k} \quad (23)$$

Step 5: Updating state estimation with the results of measurement:

$$x_{k+1/k+1} = x_{k+1/k} + K_{k+1} (y_{k+1} - C_k x_{k+1/k}) \quad (23)$$

where K is the Kalman gain matrix, and P_k is the error covariance matrix.

The program flowchart of Kalman filter is shown in the Fig. 3.

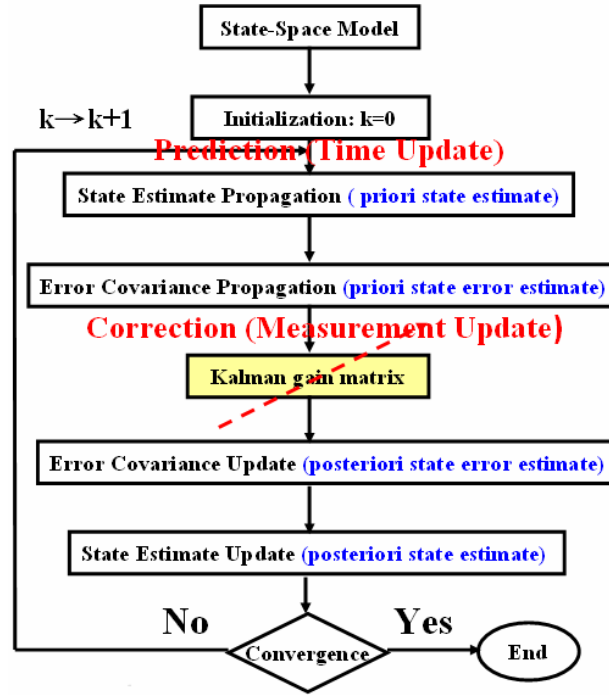


Fig. 3. The program flowchart of Kalman filter process.

C. Integrated Control Scheme

Motor electromagnetic torque and position information are used to estimate the estimated equivalent disturbance $\hat{T}_d$. Equation (13) states that inertia variation would be treated as an estimated disturbance only during acceleration and deceleration. Accordingly, the system inertia should be estimated when the motor speed changes and remain the last updated value when the motor runs at constant speed. To estimate the actual external disturbance and identify the inertia simultaneously, Fig. 4 shows the program flow to distinguish the actual external disturbance T_L and inertia-influenced disturbance $T_{\Delta J}$ from the equivalent disturbance $\hat{T}_d$ estimated by the disturbance observer.

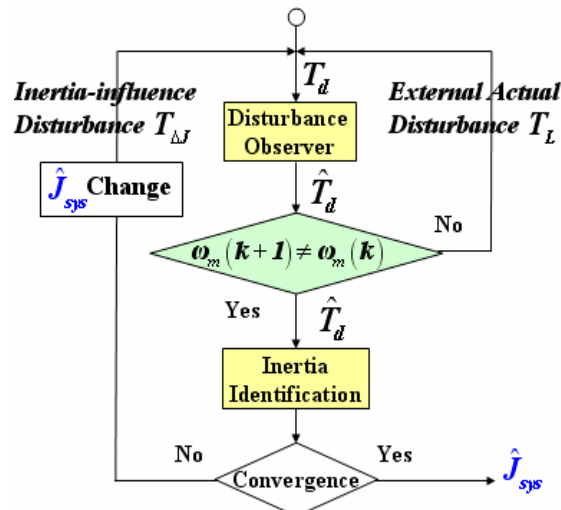


Fig. 4. The program flowchart distinguishes external actual disturbance and inertia-influence disturbance.

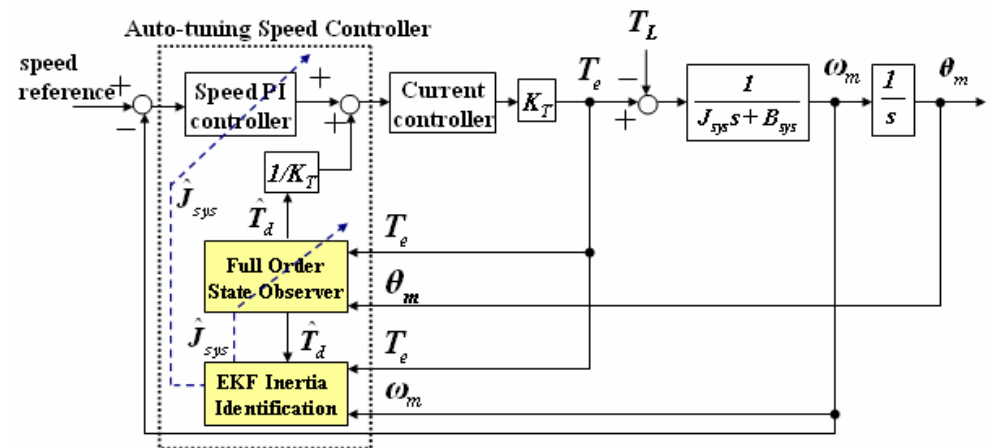


Fig.5. Block diagram of auto tuning speed controller with EKF-based inertia identification and disturbance observer.

Table 1. PARAMETERS OF PERMANENT MAGNET SYNCHRONOUS MOTOR.

Number of poles	8
Rating Power	0.2kW
Rating current	1.63A
Moment of Inertia	$0.182 \text{ kg} \cdot \text{m}^2 \times 10^{-4}$
Permitted load of Inertia	20 times motor inertia
Torque constant (K_T)	0.3907 N-m/A
Encoder Resolution	2000 pulse/rev

A block diagram of the proposed auto-tuning speed controller is shown in Fig. 5. The estimated inertia $\hat{J}_{sys}$ is used to tune the gain of the speed PI controller and to correct the reference model of the disturbance observer. Equation (24) states the gain tuning of PI controller by $\hat{J}_{sys}$.

$$\hat{J}_{sys} \cdot K_P \cdot \left(1 + \frac{K_I}{K_P \cdot s} \right) \quad (24)$$

Figure 6 shows the step response of speed controller tuned by estimated system inertia of 200%.

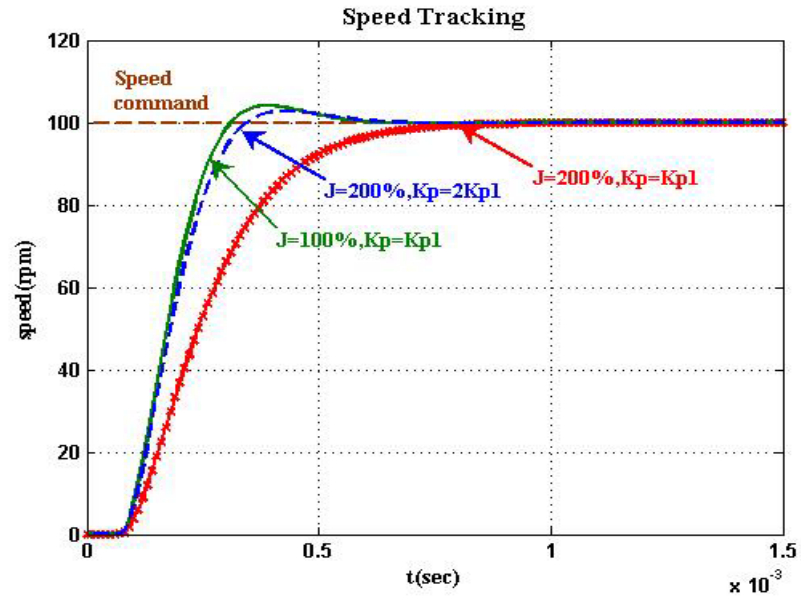


Fig.6. The step response of speed controller with estimated system inertia changed PI gain ($J=200\%$).

Furthermore, the inertia-influenced disturbance $T_{\Delta I}$ would be removed from $\hat{T}_d$ after the estimated inertia $\hat{J}_{sys}$ converges to the actual system inertia. In addition, the estimated equivalent disturbance $\hat{T}_d$ will adjust the current reference whether or not the angular speed changes. Since the estimated disturbance torque $\hat{T}_d$ is affected by the parameter variations and load effect affects, it is unlikely to distinguish each component. Therefore, direct feedback this estimated disturbance torque by attaching a gain $1/K_T$ to the current controller to compensate the effects of load and parameter variation will be effective for the robust control design, as shown in Fig. 5.

4. Simulation analysis

The results of simulations using MATLAB/Simulink confirmed the validity of the proposed algorithm. The parameters of the simulated PMSM are shown in the Table 1. The triple poles for the full-order state observer are $\beta_1=\beta_2=\beta_3=-50$. Therefore, $k_1=149.937$, $k_2=7490.628$, and $k_3=-100$. To verify that the inertia identification and load disturbance compensation are tuning simultaneously, the speed reference is alternated stepwise from 0 rpm to 400 rpm with a 1 Hz frequency. Figs. 7 to 11 show the simulation results for the case of a 300% system inertia ratio $\hat{J}_{sys}/J_m$ when an actual external disturbance $T_L=3$ N-m is applied at 0.7 sec. In Fig. 5, the step response overshoots from 0 to 4 sec. During this period, the controller calculates the load inertia ratio $\hat{J}_{sys}/J_m$, as shown in Fig. 8. Figure 9 shows the estimated equivalent disturbance $\hat{T}_d$. The estimated speed error is shown in Fig. 10, and the electromagnetic torque T_e is shown in Fig. 11. After five tuning stepwise cycles, the estimated $\hat{J}_{sys}/J_m$ approaches 300% (Fig. 8), and the overshoot of the speed response diminishes (Fig. 7). It can be seen that the estimated $\hat{J}_{sys}/J_m$ changed only when the motor was accelerating or decelerating. In the beginning, $\hat{T}_d$ and the estimated speed had errors due to the incorrect estimated system inertia $\hat{J}_{sys}$ instead of T_L . Therefore, $\hat{T}_d$ converges to zero after $\hat{J}_{sys}/J_m$ is identified at the correct value of 300%.

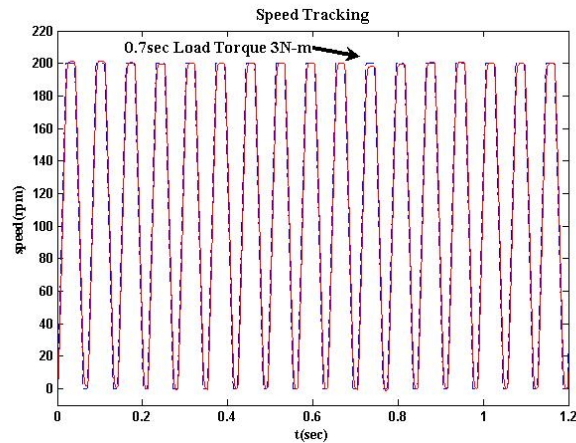


Fig. 7. Auto-tuning speed response for 300% system inertia ratio and an external actual disturbance $T_L=3$ N-m is applied at 0.7 sec.

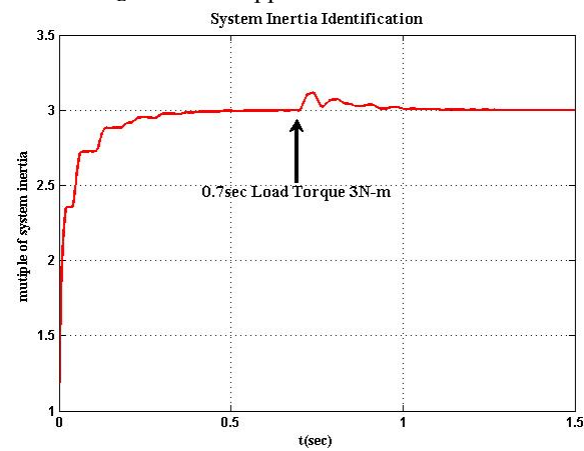


Fig. 8. Estimation of 300% system inertia ratio and an external actual load disturbance $T_L=3$ N-m is applied at 0.7 sec.

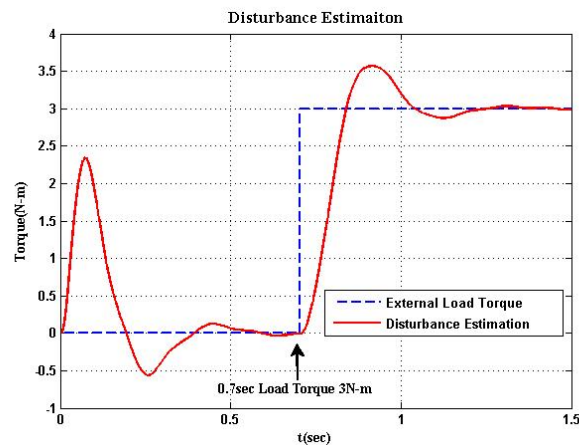


Fig. 9. Through the process of inertia identification, a starting torque and an external disturbance $T_L=3$ N-m applied at 0.7 sec for 300% system inertia ratio are estimated.



Fig. 10. Estimation of speed error for 300% system inertia ratio and an external actual disturbance $T_L=3$ N-m is applied at 0.7 sec.

The estimated speed will have errors when the estimated system inertia ratio $\hat{J}_{sys}/J_m$ is not tuned suitably and the actual external disturbance is not estimated. When the actual external disturbance $T_L=3$ N-m is applied at 0.7 sec, the estimated equivalent disturbance $\hat{T}_d$ will rise to the accurate actual disturbance value of 3 N-m, revealing that the inertia-influenced disturbance T_{Δ} will be removed from $\hat{T}_d$. Another effect of the disturbance, shown in Fig. 10, is that the speed response decreased to about 2.5 rpm and was regulated back to 400 rpm after four tuning stepwise cycles.

Fig. 11 shows the electromagnetic torque T_e compensation when the 3 N-m actual external disturbance T_L is applied. However, T_e will not compensate for the inertia-influenced disturbance T_{Δ} when the motor starts running, as shown in Figs. 9 and 11.

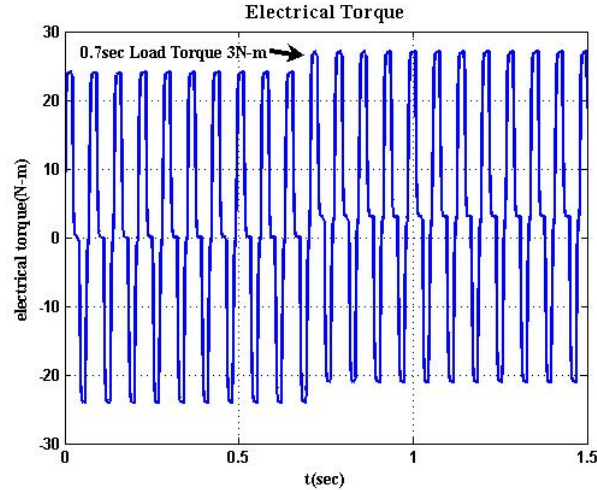


Fig. 11. Electromagnetic torque T_e for 300% system inertia ratio and an external actual disturbance $T_L=3$ N-m is applied at 0.7 sec. The electromagnetic torque is generated very quick to adapt to the applied load.

5. Experiment results

A. Experimental Setup

The configuration of the experimental servo motor driver system is shown in Fig 12. The proposed control scheme is implemented with DSP TMS320F2812 and a pulse-width modulation (PWM) inverter composed of insulated-gate bipolar transistors (IGBTs). Fig. 13 shows the experimental setup, in which the PMSM is coupled to a DC generator through the transmission shaft. The power output of the generator is connected to the DC electrical load. Accordingly, the current load variation of the DC electrical load is equivalent to applying an

actual external disturbance T_L to the PMSM. All simulations and experiments are performed with the parameters listed in Table 1.

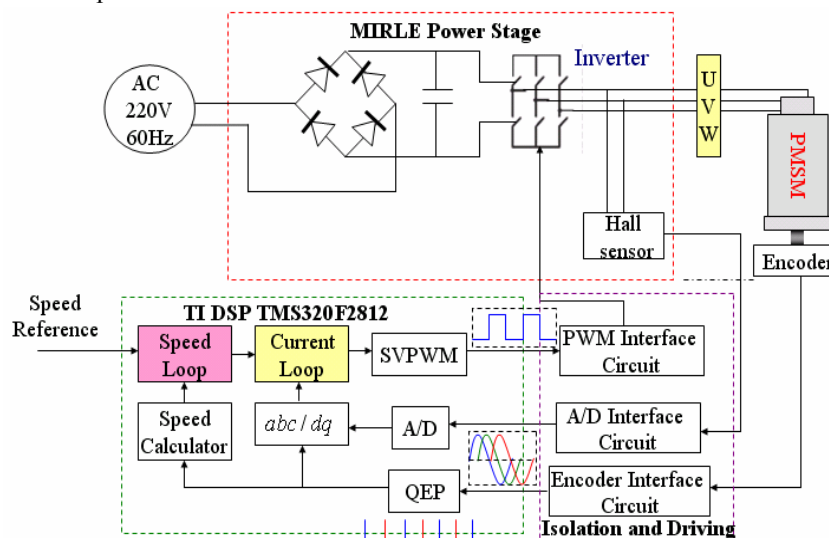


Fig. 12. PMSM servo motor driver system.

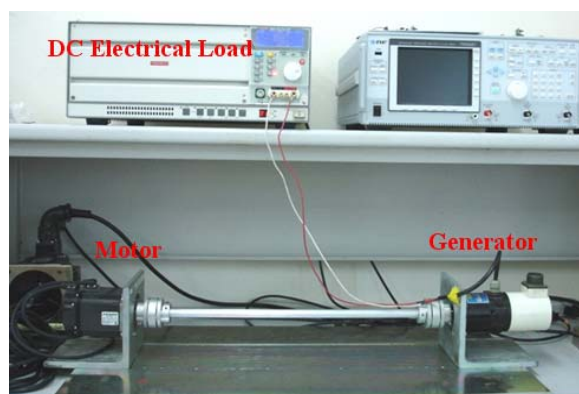
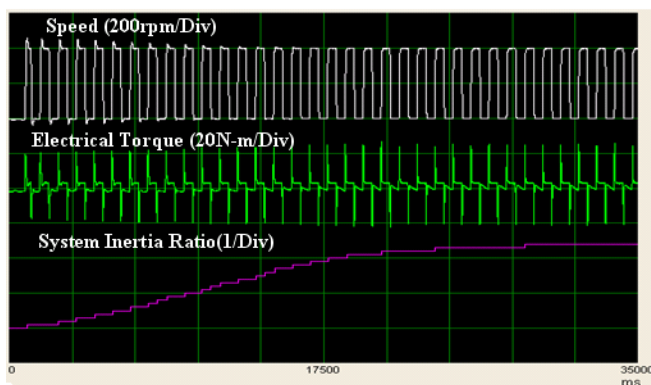


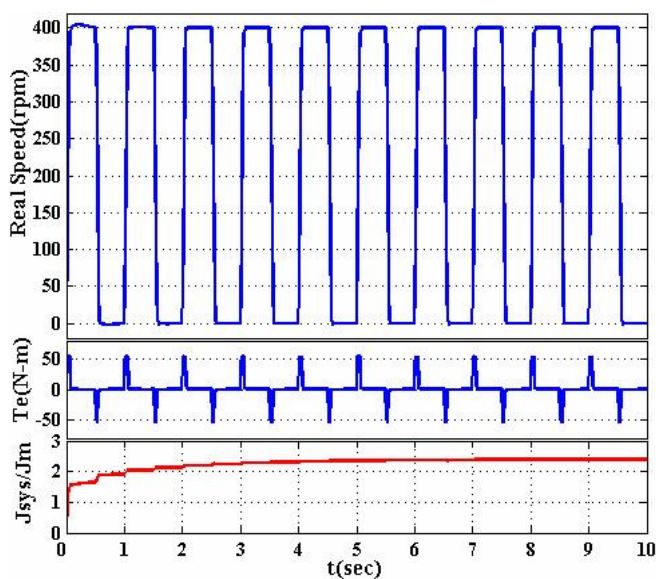
Fig. 13. The experimental setup. A DC electrical load is applied to generator to simulate the external load disturbances.

B. Experimental Results

Model reference adaptive control algorithms⁽¹⁶⁾⁽²¹⁾ are typically used in industrial controllers to auto-tune the speed performance and suppress vibration. We compared the proposed auto-tuning speed controller with a PMSM industrial controller, model MR-J3 which is an AC servo drive made by Mitsubishi, in real-time experiments under different practical disturbance and inertia variation conditions. Figs. 14 and 15 show the experimental results for the industrial and proposed controllers, respectively. The speed reference is alternated stepwise from 0 rpm to 400 rpm with 1 Hz frequency. Results are similar to the simulation results. The real speed response exhibited undesirable overshoot if the estimated system inertia ratio $\hat{J}_{sys}/J_m$ did not tune suitably. The system inertia ratio of the industrial controller is identified as 240% of the correct value after 29 tuning stepwise cycles [Fig. 14(a)]. However, the proposed EKF-based controller needs only 6 tuning stepwise cycles to identify the system inertia ratio correctly [Fig. 14(b)]. For 400% load inertia identification, 20 cycles (13.5 sec) are needed for MR-J3, as shown in Fig. 15(a). Our EKF-based algorithm controller will take 12 cycles (8 sec) to measure this 400% load inertia [Fig. 15(b)].

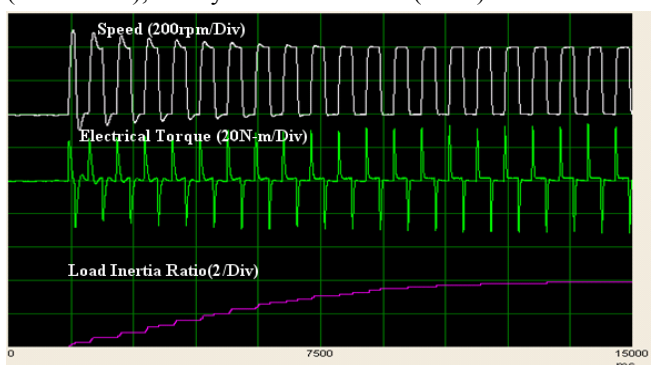


(a) MR-J3 inertia identification response for 240% load inertia. It takes about 31.5 sec.

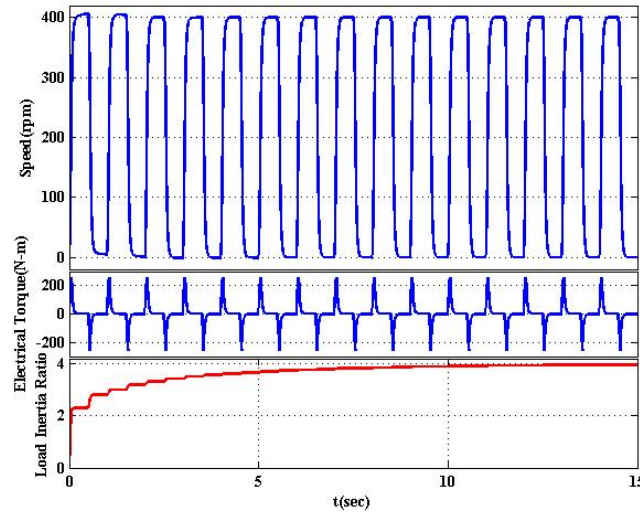


(b) Proposed controller inertia identification response for 240% load inertia.

Fig. 14. Speed response of proposed controller for 240% system inertia ratio without an external actual disturbance T_L . From top: real speed (50rpm/div), electromagnetic torque T_e (50N-m/div), and system inertia ratio (1/div).



(a) MR-J3 response for 400% load inertia. It takes about 13.5 sec.



(b) Proposed controller response for 400% load inertia.

Fig. 15. Speed response of proposed controller for 400% system inertia ratio without an external actual disturbance T_L . From top: real speed (100 rpm/div), electromagnetic torque T_e (200 N-m/div), and system inertia ratio (2/div).

Figs. 16 and 17 show the experimental results for the industrial and proposed controllers, respectively, for 240% system inertia and an actual external disturbance of $T_L=20$ N-m. In Fig. 16, the estimated system inertia ratio $\hat{J}_{sys}/J_m$ of the industrial PMSM controller is obviously influenced by T_L , decreasing from 240% to 200%. An incorrect $\hat{J}_{sys}/J_m$ will degrade the speed performance accordingly. In Fig. 15, although $\hat{J}_{sys}/J_m$ of the proposed auto-tuning controller is also influenced by T_L , $\hat{J}_{sys}/J_m$ converged precisely to 240% after eight tuning stepwise cycles, revealing that the inertia-influenced disturbance T_{Δ} is removed from the estimated equivalent disturbance $\hat{T}_d$. By fast estimation of the EKF, the system inertia ratio can be identified quickly. In addition, the real speed is also regulated when T_L is applied at 10 sec. $\hat{T}_d$ of the proposed controller is fed forward to adjust the electromagnetic torque T_e , as shown in Fig. 17.

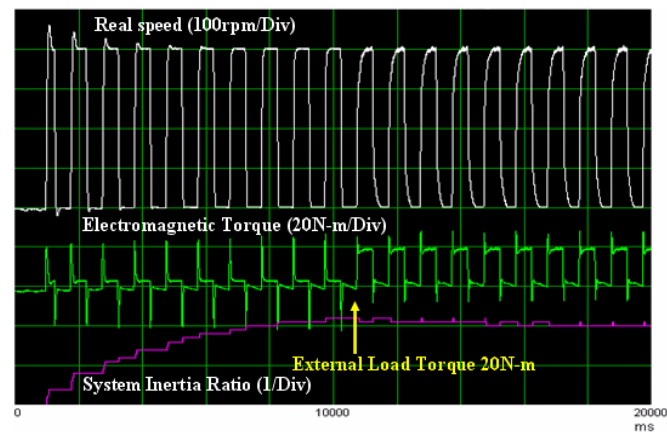


Fig. 16. Speed response of MR-J3 industrial PMSM controller for 240% system inertia ratio and an external actual disturbance $T_L=20$ N-m is applied at 11sec. From top: real speed (100 rpm/div), electromagnetic torque T_e (20 N-m/div), and system inertia ratio (1/div).

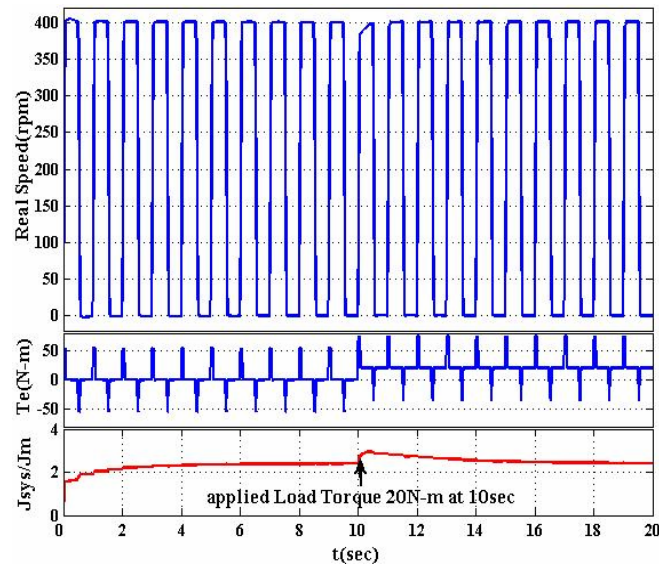


Fig. 17. Speed response of proposed controller for 240% system inertia ratio and an external actual disturbance $T_L=20$ N-m is applied at 10sec. From top: real speed (50 rpm/div), electromagnetic torque T_e (50 N-m/div), and system inertia ratio (2/div).

6. Conclusions

This paper has presented an on-line auto-tuning control scheme based on EKF inertia identification and a full-order state observer to tune the speed controller, to compensate for an external disturbance, and to adjust the observer of the system model. The simulation and experimental results confirmed that system inertia identification and disturbance estimation can be executed simultaneously. The proposed algorithm can distinguish the source of the estimated disturbance from the variation of in electromagnetic torque. Moreover, this paper verified that the system inertia is accurately found by observing the estimated equivalent disturbance and the differential of the speed feedback. The proposed estimation scheme and auto-tuning control algorithm is also compared with the an industrial PMSM controller in real-time experiments for different practical conditions to demonstrate the validity and effectiveness of faster identification capability and robust characteristics.

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